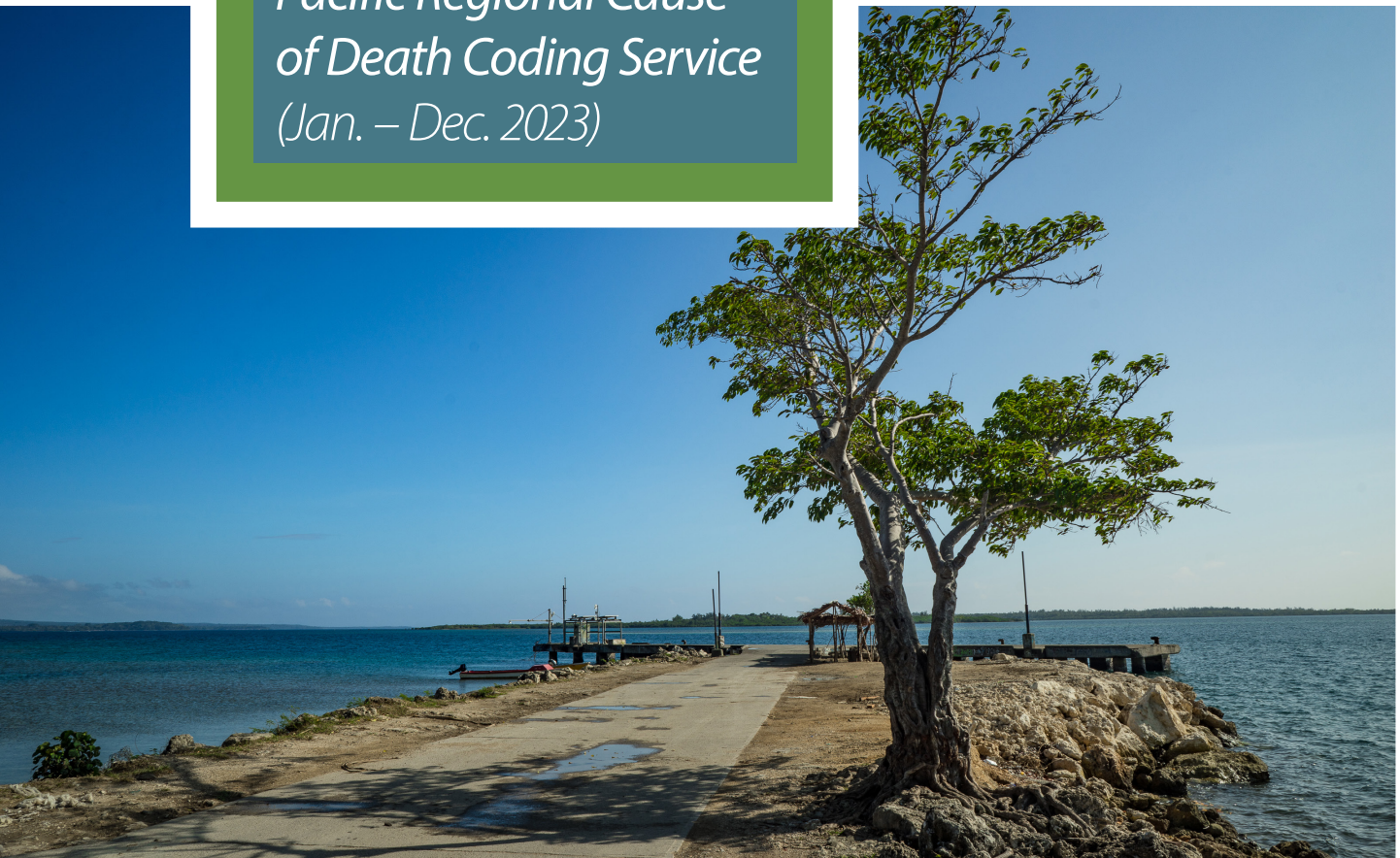




Pacific
Community
Communauté
du Pacifique

Making Death Count

***Vanuatu: Causes of
Death Analysis from the
Pacific Regional Cause
of Death Coding Service
(Jan. – Dec. 2023)***



WORLD BANK GROUP

OVERVIEW

The report summarises key mortality patterns by month, cause of death, sex, age group and place of death. It also highlights the leading causes of death, the burden of non-communicable diseases, and the extent of ill-defined causes, which are important indicators of death certification and coding quality.

METHODOLOGY

This report presents an analysis of 2023 mortality data submitted by the Vanuatu Bureau of Statistics (VBOS) to the Pacific Regional Cause of Death Coding Service (RCS). The dataset includes only deaths recorded through Vanuatu National Hospital and primarily covers deaths in Port Vila and surrounding areas, where families are generally required to obtain a hospital death certificate before burial at the municipal site. Medical certificates of cause of death (MCCDs) are not routinely issued for all deaths—only 3% in 2024—contributing to the low number of medically certified deaths. The dataset excludes deaths from rural Efate and other provincial hospitals in Vanuatu. As a result, the number of records is substantially lower than the national number of deaths estimated from the 2020 census crude death rate.

A total of 517 de-identified death records were submitted, including 33 stillbirths. The stillbirths were removed from the dataset and were not included in this analysis. This represents 20.7% of Vanuatu’s expected annual deaths, based on an estimated 2327 deaths per year and a crude death rate of 7.8 per 1,000 population using the 2020 census report.

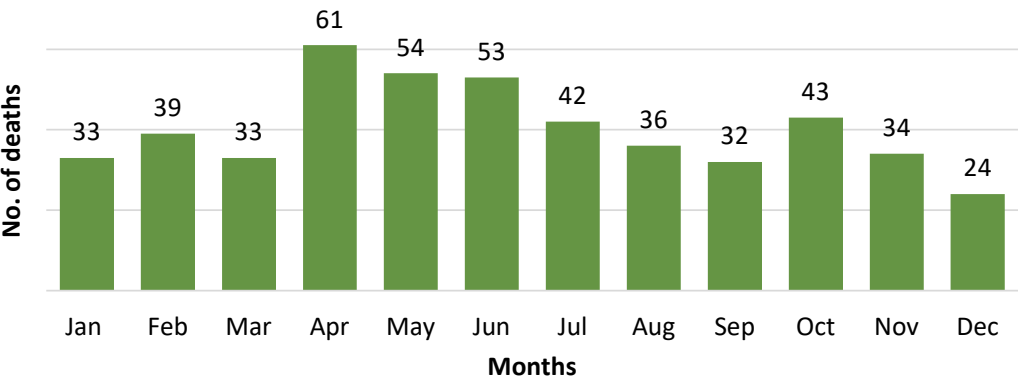
The analysis is based on the underlying cause of death selected by Iris, the automated mortality coding software used by the RCS. The causes of death are coded using the WHO statistical classification system, the International Statistical Classification of Diseases and Related Health Problems 10th revision (ICD-10). The findings are intended to support interpretation of mortality trends to inform public health interventions and lead to ongoing improvements in death registration, certification and cause-of-death coding in Vanuatu.

RESULTS

Death records received for each month

The monthly distribution of deaths in 2023 shows variation across the year. This pattern helps identify possible seasonal trends, reporting fluctuations, or months that may require further review to better understand mortality and registration patterns.

Figure 1. Number of deaths by month



Further review of the monthly figures would help confirm whether peaks reflect seasonal mortality, delayed registration, or reporting variation.

Top 10 causes of death by ICD-10 Chapter level

Table 1 show that deaths were highly concentrated among a small number of ICD-10-chapter groups, with the top 10 causes accounting for 459 deaths (94.8%) out of 484 deaths recorded in 2023. Diseases of the circulatory system were the leading cause, contributing 125 deaths (25.8%), followed by endocrine, nutritional and metabolic diseases with 83 deaths (17.1%) and neoplasms with 62 deaths (12.8%). Together, these leading groups highlight the major burden of non-communicable diseases in the submitted mortality data. Ill-defined causes ranked fourth, accounting for 47 deaths (9.7%), indicating that improved death certification and more specific cause-of-death information remain important priorities for strengthening mortality statistics.

Table 1. Top 10 causes of death by ICD-10-chapter level

Rank	ICD-10 code	Diseases	No. of deaths	Percentage (%)
1	I00-I99	Diseases of the circulatory system	125	25.8
2	E00-E90	Endocrine, nutritional and metabolic diseases	83	17.1
3	C00-D48	Neoplasms	62	12.8
4	R00-R99	Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified	47	9.7
5	A00-B99	Certain infectious and parasitic diseases	45	9.3
6	J00-J99	Diseases of the respiratory system	31	6.4
7	N00-N99	Diseases of the genitourinary system	24	5.0
8	K00-K93	Diseases of the digestive system	15	3.1
9	V01-Y98	External causes of morbidity and mortality	14	2.9
10	G00-G99	Diseases of the nervous system	13	2.7
	L00-L99, P00-P96, M00-M99, D50-D89, Q00-Q99, F00-F99	Remainder of other chapters	25	5.2
Total 2,023 deaths			484	100.0

Top 5 causes of death by sex (ICD-10-chapter level)

Table 2 show that the leading causes of death were broadly similar for males and females, with non-communicable diseases dominating both groups. Among males, diseases of the circulatory system were the leading cause, accounting for 88 deaths (29.9%), followed by endocrine, nutritional and metabolic diseases with 49 deaths (16.7%). Among females, circulatory diseases and neoplasms were equal leading causes, each accounting for 37 deaths (19.4%), followed by endocrine, nutritional and metabolic diseases with 34 deaths (17.8%). Ill-defined causes were more common among males, with 32 deaths (10.9%) compared with 15 deaths (7.9%) among females, suggesting a greater need to improve cause-of-death specificity for male deaths. Overall, the top five causes accounted for more than 80% of deaths in both sexes, indicating that mortality was concentrated in a small number of major cause groups.

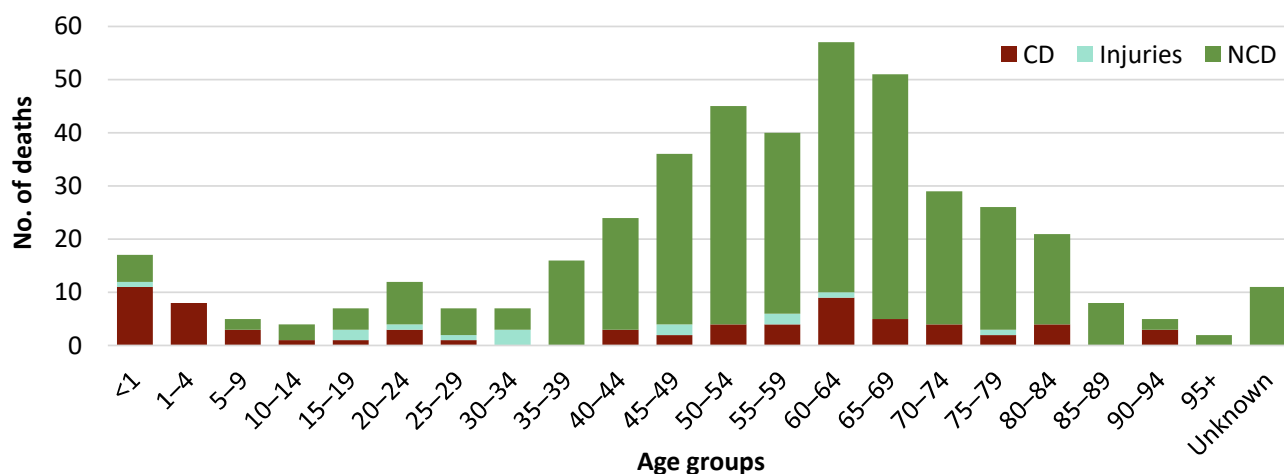
Table 2. Top 5 causes of mortality by sex

Rank	ICD-10 Code	Female			Rank	ICD-10 Code	Male		
		Diseases	No. of deaths	Percentage (%)			Diseases	No. of deaths	Percentage (%)
1=	I00-I99	Diseases of the circulatory system	37	19.5	1	I00-I99	Diseases of the circulatory system	88	29.9
1=	C00-D48	Neoplasms	37	19.5	2	E00-E90	Endocrine, nutritional and metabolic diseases	49	16.7
2	E00-E90	Endocrine, nutritional and metabolic diseases	34	17.9	3	R00-R99	Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified	32	10.9
3	A00-B99	Certain infectious and parasitic diseases	20	10.5	4=	A00-B99	Certain infectious and parasitic diseases	25	8.5
4	R00-R99	Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified	15	7.9	4=	C00-D48	Neoplasms	25	8.5
5	J00-J99	Diseases of the respiratory system	11	5.8	5	J00-J99	Diseases of the respiratory system	20	6.8
		Total (top 5 causes)	154	81.1			Total (top 5 causes)	239	81.3
		Total female deaths	190				Total male deaths	294	

Non-Communicable, Communicable and Injuries deaths

Figure 2 show that non-communicable diseases were the dominant cause of death across the age groups, particularly among adults and older age groups, reflecting the major burden of circulatory diseases, diabetes-related conditions and cancers. Communicable disease deaths were fewer overall but remained present across some age groups, while injury-related deaths were the smallest contributor, with external causes accounting for 14 deaths (2.9%). The pattern highlights the importance of strengthening NCD prevention, early diagnosis and long-term disease management, while also maintaining targeted action on communicable diseases and injury prevention.

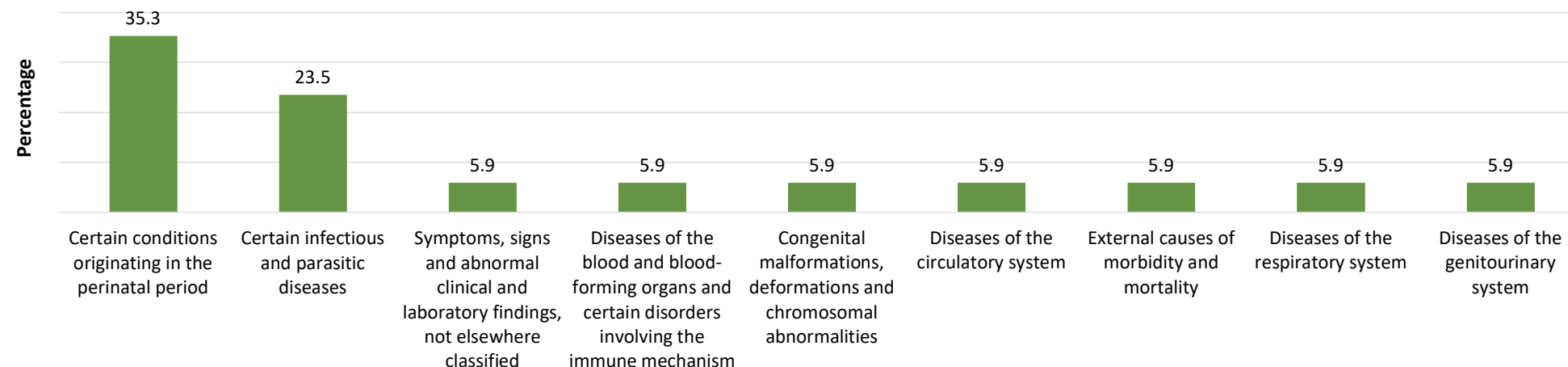
Figure 2. Number of deaths due to NCD, CD and Injuries by age groups



Under-1 causes of death

Figure 3 show that under-1 deaths were mainly associated with conditions originating in the perinatal period, which accounted for 6 deaths. The leading subcategories were perinatal infections and newborns affected by maternal factors or complications of pregnancy, labour and delivery, each accounting for 2 deaths. Although the number of under-1 deaths was small, the pattern highlights the importance of strengthening antenatal care, safe delivery practices, newborn infection prevention and early neonatal care to reduce preventable infant deaths.

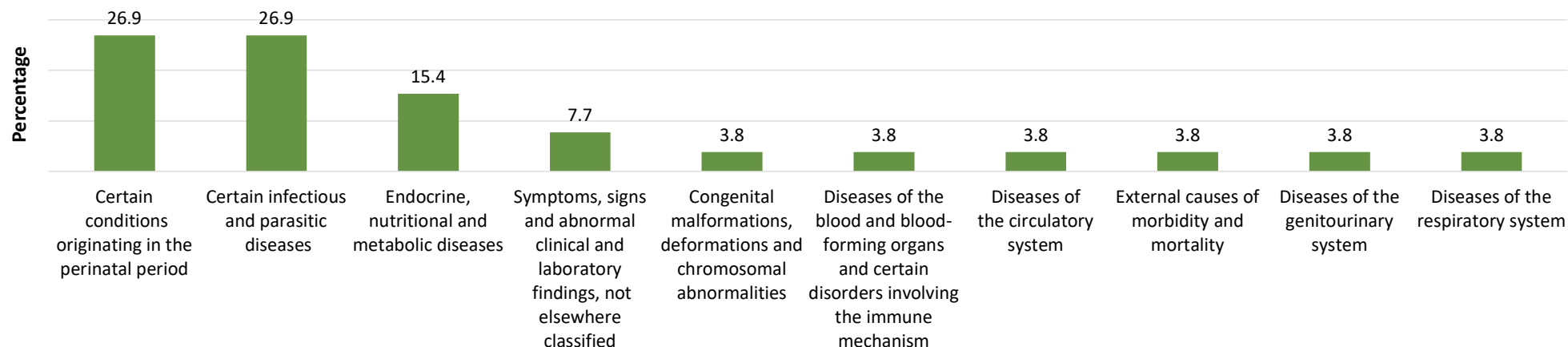
Figure 3. Under 1 cause of death



Under 5 causes of death

Figure 4 shows that under-5 deaths were mainly associated with conditions originating in the perinatal period and certain infectious and parasitic diseases, each accounting for 7 deaths. This suggests that mortality among children under five was concentrated around early-life conditions, including maternal, delivery and newborn-related factors, as well as infectious causes. Although the number of under-5 deaths was small, the pattern highlights the importance of strengthening antenatal care, safe delivery practices, newborn infection prevention, early neonatal care and follow-up for high-risk infants.

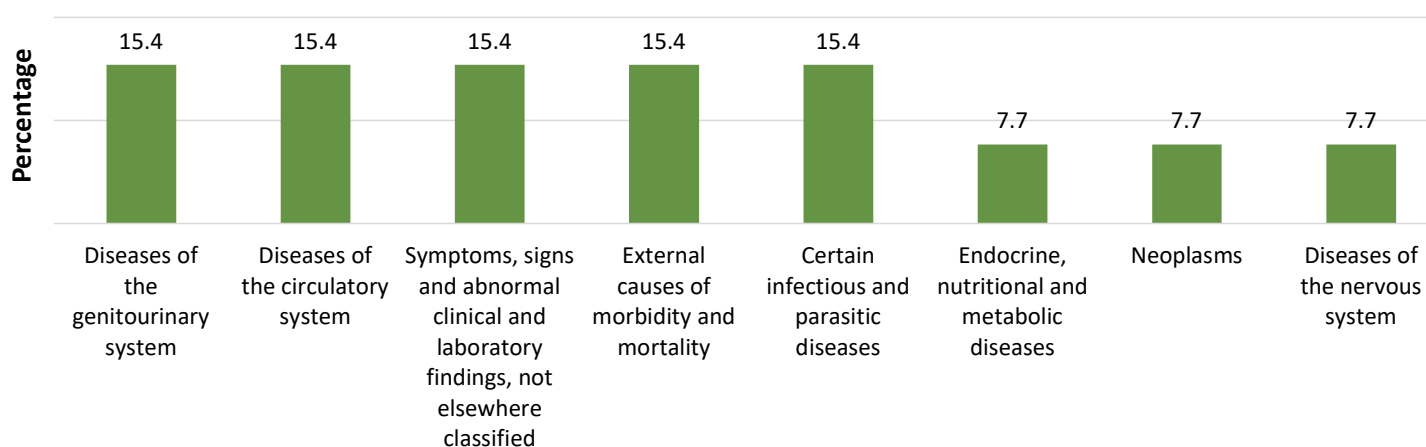
Figure 4. Under 5 causes of death



Teenage deaths

Teenage deaths were spread across several ICD-10 cause groups, with no single cause group dominating the pattern. Five cause groups each recorded 2 deaths, representing 15.4% of teenage deaths respectively: diseases of the genitourinary system, which may include kidney and urinary system conditions; diseases of the circulatory system, including heart and vascular conditions; symptoms, signs and ill-defined causes, where the underlying cause was not clearly specified; external causes, such as injuries and other non-disease events; and certain infectious and parasitic diseases. A further three cause groups each recorded 1 death (7.7%): endocrine, nutritional and metabolic diseases, which include diabetes and related metabolic conditions; neoplasms; and diseases of the nervous system. Although the number of teenage deaths was small, the distribution shows that mortality in this age group was not concentrated in one cause category, but reflected a mix of chronic conditions, infectious causes, injuries, and less specific or ill-defined causes. These findings should therefore be interpreted cautiously and may warrant further case-level review to better understand preventable factors and improve cause-of-death specificity.

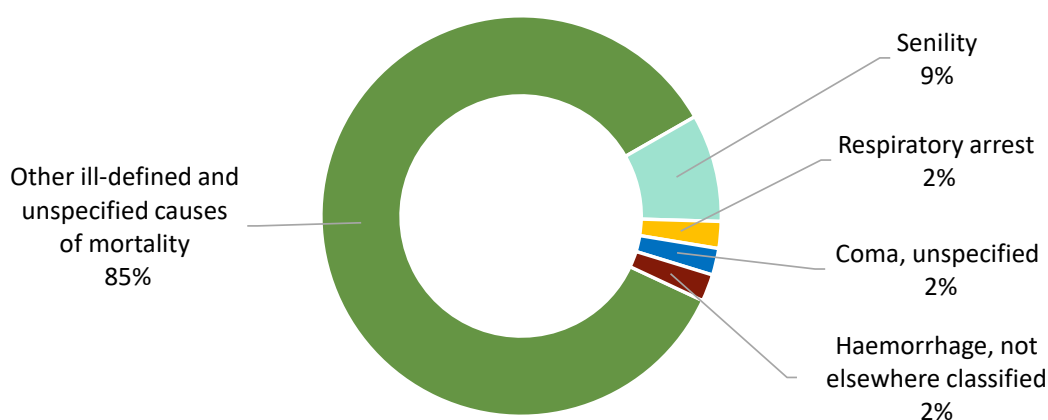
Figure 5. Teenage causes of death



Ill-defined causes of death

Ill-defined causes (R00-R99) accounted for 47 deaths (9.7%). Most were coded as ill-defined and unknown causes of mortality, with 40 deaths (85% of ill-defined deaths), highlighting the need for more specific death certification.

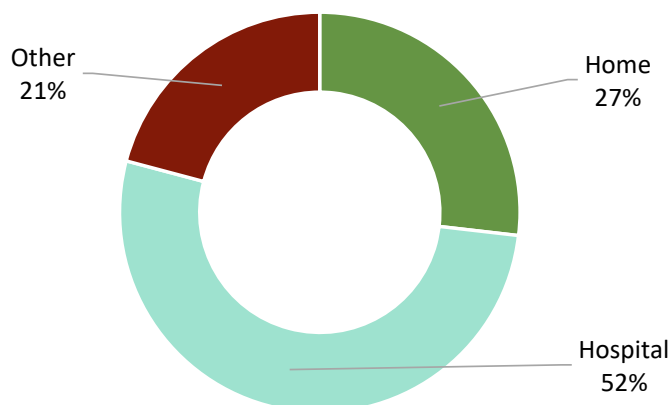
Figure 6. Ill-defined causes of death



DEATHS BY PLACE OF DEATH

The place-of-death distribution shows that most deaths were recorded in hospitals, with 253 deaths, representing approximately 52.3% of deaths with a recorded place of death. Home deaths accounted for 130 deaths (26.9%), while 101 deaths (20.9%) were recorded in other locations. This pattern indicates that while hospitals remain the main setting for recorded deaths, a substantial proportion occurred outside hospital settings, highlighting the importance of strengthening community death notification, timely certification and registration processes.

Figure 7. Percentage of deaths by place of death



Causes of death by three-character code level

At the three-character ICD-10 code level, deaths were concentrated in a small number of major cause groups. Diseases of the circulatory system were the leading cause, accounting for 125 deaths (25.8%), with ischaemic heart diseases, cerebrovascular diseases and hypertensive diseases contributing the largest shares within this group. Endocrine, nutritional and metabolic diseases were the second leading group, with 83 deaths (17.1%), driven mainly by diabetes mellitus. Neoplasms accounted for 62 deaths (12.8%), with female genital, breast and digestive organ cancers among the main subcategories. Ill-defined causes accounted for 47 deaths (9.7%), most of which were coded as ill-defined and unknown causes of mortality, highlighting the need to improve cause-of-death certification and reduce non-specific coding. Infectious and parasitic diseases, respiratory diseases, genitourinary diseases and external causes also contributed to the overall mortality pattern.

Table 3. Causes of death by three-character code level

ICD-10 Chapter Code	Disease	Number of deaths	Percentage (%)
I00-I99	Diseases of the circulatory system	125	25.8
I20-I25	Ischaemic heart diseases	59	47.2
I60-I69	Cerebrovascular diseases	28	22.4
I10-I15	Hypertensive diseases	23	18.4
I30-I52	Other forms of heart disease	10	8.0
I05-I09	Chronic rheumatic heart diseases	5	4.0
E00-E90	Endocrine, nutritional and metabolic diseases	83	17.1
E10-E14	Diabetes mellitus	71	85.5
E70-E90	Metabolic disorders	7	8.4
E40-E46	Malnutrition	4	4.8
E00-E07	Disorders of thyroid gland	1	1.2

ICD-10 Chapter Code	Disease	Number of deaths	Percentage (%)
C00-D48	Neoplasms	62	12.8
C51-C58	Malignant neoplasms of female genital organs	13	21.0
C50-C50	Malignant neoplasm of breast	13	21.0
C15-C26	Malignant neoplasms of digestive organs	12	19.4
C76-C80	Malignant neoplasms of ill-defined, secondary and unspecified sites	8	12.9
C30-C39	Malignant neoplasms of respiratory and intrathoracic organs	4	6.5
C81-C96	Malignant neoplasms, stated or presumed to be primary, of lymphoid, haematopoietic and related tissue	3	4.8
C64-C68	Malignant neoplasms of urinary tract	3	4.8
C45-C49	Malignant neoplasms of mesothelial and soft tissue	1	1.6
D37-D48	Neoplasms of uncertain or unknown behaviour	1	1.6
D10-D36	Benign neoplasms	1	1.6
C73-C75	Malignant neoplasms of thyroid and other endocrine glands	1	1.6
C69-C72	Malignant neoplasms of eye, brain and other parts of central nervous system	1	1.6
C60-C63	Malignant neoplasms of male genital organs	1	1.6
R00-R99	Symptoms, signs and abnormal clinical and laboratory findings, not elsewhere classified	47	9.7
R95-R99	Ill-defined and unknown causes of mortality	40	85.1
R50-R69	General symptoms and signs	5	10.6
R00-R09	Symptoms and signs involving the circulatory and respiratory systems	1	2.1
R40-R46	Symptoms and signs involving cognition, perception, emotional state and behaviour	1	2.1
A00-B99	Certain infectious and parasitic diseases	45	9.3
A30-A49	Other bacterial diseases	26	57.8
B15-B19	Viral hepatitis	6	13.3
A15-A19	Tuberculosis	5	11.1
A00-A09	Intestinal infectious diseases	5	11.1
A20-A28	Certain zoonotic bacterial diseases	1	2.2
B99-B99	Other infectious diseases	1	2.2
B20-B24	Human immunodeficiency virus [HIV] disease	1	2.2
J00-J99	Diseases of the respiratory system	31	6.4
J40-J47	Chronic lower respiratory diseases	16	51.6
J09-J18	Influenza and pneumonia	9	29.0
J20-J22	Other acute lower respiratory infections	2	6.5
J95-J99	Other diseases of the respiratory system	1	3.2
J90-J94	Other diseases of pleura	1	3.2
J80-J84	Other respiratory diseases principally affecting the interstitium	1	3.2
J60-J70	Lung diseases due to external agents	1	3.2
N00-N99	Diseases of the genitourinary system	24	5.0
N17-N19	Renal failure	18	75.0
N40-N51	Diseases of male genital organs	3	12.5
N25-N29	Other disorders of kidney and ureter	2	8.3
N20-N23	Urolithiasis	1	4.2
K00-K93	Diseases of the digestive system	15	3.1
K70-K77	Diseases of liver	7	46.7
K55-K64	Other diseases of intestines	4	26.7
K90-K93	Other diseases of the digestive system	1	6.7
K80-K87	Disorders of gallbladder, biliary tract and pancreas	1	6.7
K20-K31	Diseases of oesophagus, stomach and duodenum	1	6.7

ICD-10 Chapter Code	Disease	Number of deaths	Percentage (%)
K35-K38	Diseases of appendix	1	6.7
V01-Y98	External causes of morbidity and mortality	14	2.9
V01-V99	Transport accidents	5	35.7
X58-X59	Accidental exposure to other and unspecified factors	4	28.6
X60-X84	Intentional self-harm	1	7.1
W00-W19	Falls	1	7.1
Y10-Y34	Event of undetermined intent	1	7.1
W65-W74	Accidental drowning and submersion	1	7.1
X00-X09	Exposure to smoke, fire and flames	1	7.1
G00-G99	Diseases of the nervous system	13	2.7
G90-G99	Other disorders of the nervous system	5	38.5
G80-G83	Cerebral palsy and other paralytic syndromes	3	23.1
G40-G47	Episodic and paroxysmal disorders	2	15.4
G60-G64	Polyneuropathies and other disorders of the peripheral nervous system	1	7.7
G20-G26	Extrapyramidal and movement disorders	1	7.7
G30-G32	Other degenerative diseases of the nervous system	1	7.7
L00-L99	Diseases of the skin and subcutaneous tissue	9	1.9
L80-L99	Other disorders of the skin and subcutaneous tissue	6	66.7
L00-L08	Infections of the skin and subcutaneous tissue	3	33.3
P00-P96	Certain conditions originating in the perinatal period	7	1.4
P35-P39	Infections specific to the perinatal period	2	28.6
P00-P04	Fetus and newborn affected by maternal factors and by complications of pregnancy, labour and delivery	2	28.6
P50-P61	Haemorrhagic and haematological disorders of fetus and newborn	1	14.3
P05-P08	Disorders related to length of gestation and fetal growth	1	14.3
P20-P29	Respiratory and cardiovascular disorders specific to the perinatal period	1	14.3
M00-M99	Diseases of the musculoskeletal system and connective tissue	5	1.0
M70-M79	Other soft tissue disorders	2	40.0
M45-M49	Spondylopathies	1	20.0
M15-M19	Arthrosis	1	20.0
M30-M36	Systemic connective tissue disorders	1	20.0
D50-D89	Diseases of the blood and blood-forming organs and certain disorders involving the immune mechanism	2	0.4
D65-D69	Coagulation defects, purpura and other haemorrhagic conditions	1	50.0
D50-D53	Nutritional anaemias	1	50.0
Q00-Q99	Congenital malformations, deformations and chromosomal abnormalities	1	0.2
Q20-Q28	Congenital malformations of the circulatory system	1	100.0
F00-F99	Mental and behavioural disorders	1	0.2
F50-F59	Behavioural syndromes associated with physiological disturbances and physical factors	1	100.0
	Total	484	

KEY FINDINGS

- The dataset contains only deaths recorded through the Vanuatu National Hospital and represents only an estimated 20.7% of deaths. Deaths varied by month, suggesting possible seasonal trends, reporting fluctuations, or months requiring further review.

- The **top 10 causes** accounted for **459 deaths (94.8%)**, showing that deaths were concentrated in a small number of major cause groups.
- **Circulatory diseases** were the leading cause of death, with **125 deaths (25.8%)**.
- **Endocrine, nutritional and metabolic diseases** were second, with **83 deaths (17.1%)**, driven mainly by **diabetes mellitus**.
- **Neoplasms** were third, with **62 deaths (12.8%)**.
- **Ill-defined causes** accounted for **47 deaths (9.7%)**, highlighting the need to strengthen death certification and coding quality.
- Non-communicable diseases were the main contributor to mortality, while injuries were the smallest contributor, with **14 deaths (2.9%)**.
- Among males, circulatory diseases were the leading cause; among females, circulatory diseases and neoplasms were equal leading causes.
- Under-1 and under-5 deaths were mainly linked to **perinatal conditions**, which accounted for **7 deaths**.
- Just over half of recorded deaths occurred in hospitals (**253 deaths; 52.3%**), while **130 deaths (26.9%)** occurred at home and **101 deaths (20.9%)** in other locations, highlighting the need to strengthen notification, certification and registration for deaths outside health facilities.

KEY RECOMMENDATIONS

Based on the findings, the following actions are recommended:

- **Improve death notification and registration.** Strengthen reporting from hospitals, health facilities and communities, ensuring deaths outside facilities are promptly notified, certified and registered, and improve overall completeness of mortality reporting.
- **Strengthen prevention and management of non-communicable diseases.** Prioritise action on circulatory diseases, diabetes and cancers through improved screening, early diagnosis, regular monitoring, treatment adherence, referral pathways for high-risk patients and community awareness.
- **Improve medical certification of cause of death.** Continue training doctors and certifiers on accurate completion of medical certificates, with emphasis on recording the causal sequence of death and avoiding non-specific terms where more precise clinical information is available.
- **Review ill-defined deaths.** Conduct regular review of deaths coded to R00–R99, particularly ill-defined and unknown causes, to identify whether additional clinical or registration information can improve coding specificity.
- **Strengthen maternal, newborn and perinatal care.** Review under-1 and under-5 deaths linked to perinatal conditions to identify preventable factors, including maternal complications, delivery-related risks, newborn infections and access to timely care.
- **Use findings for public health planning.** Share the results with health and civil registration stakeholders to guide priority setting, support NCD programmes, improve data quality, and strengthen routine mortality surveillance.

LIMITATIONS

- The submitted records represent only 20.7% of expected annual deaths in Vanuatu and may also be affected by under-registration, delayed reporting or incomplete data submission; therefore, the findings describe mortality patterns in the submitted dataset and should not be used as population-level estimates without further assessment of data completeness and representativeness.
- The dataset did not include dates of birth, which limited the ability to conduct detailed age-based analysis.

- The presence of ill-defined causes indicates that some death certificates lack sufficient clinical detail, which affects the accuracy of cause-of-death coding.
- Small numbers in some categories, particularly under-1, under-5 and injury-related deaths, should be interpreted cautiously.

CONCLUSION

The 2023 mortality data submitted to the Pacific Regional Coding Service provides useful insight into leading causes of death and key mortality patterns for deaths recorded through the Vanuatu National Hospital. The findings show a strong burden of non-communicable diseases, particularly circulatory diseases, diabetes-related conditions and cancers. Ill-defined causes also remain an important data quality concern, highlighting the need to strengthen medical certification and cause-of-death coding.

Overall, the report reinforces the importance of improving completeness, timeliness and quality of mortality reporting. Strengthening death notification, certification and registration processes will support more accurate national mortality statistics and provide stronger evidence for health planning, NCD prevention, maternal and newborn care, and routine mortality surveillance.

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Noumea, New Caledonia
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